SHARPE'S THEOREM ON LINEAR EQUATIONS, AND ITS UNIT FORMALISATION

Let R be a commutative ring and F an R-module. We consider a linear system with coefficients in R and unknowns in F,

$$x_1U_1 + \cdots + x_nU_n = V, \tag{1}$$

where $U_1, \ldots, U_n$ are column vectors of R^m and $V \in F^m$. Let $A = (U_1 \cdots U_n)$ and $B = (U_1 \cdots U_n\, V)$, and write $\Delta_l(M)$ for the ideal (or, for B, the submodule of F) generated by the $l \times l$ minors of M. If $\Delta_n(A)$ is regular on F, (1) has at most one solution, and a solution forces $\Delta_{n+1}(B) = 0$. Sharpe [5] proved that grade 2 gives existence.

Theorem 1 (Sharpe)**.** *If* $\mathrm{Gr}(\Delta_n(A); F) \geq 2$ *and* $\Delta_{n+1}(B) = 0$*, the system (1) has exactly one solution. Conversely, if (1) has a solution, then* $\Delta_{n+1}(B) = 0$*.*

Grade 1 does not suffice: over $\mathbb{Z}$ the system $2x = 3$, $4x = 6$ satisfies $\Delta_2(B) = 0$, but the grade of $\langle 2, 4\rangle$ is only 1. The theorem is Theorem B-7.5 of [1], where it is derived from a local-global principle for coverings in depth 2.

The proof below is the one of the note [2]: the minors are coordinates in the exterior algebra, Cramer's rule is applied to each $n \times n$ subsystem, and the depth-2 hypothesis is used once, on the Cramer numerators, which are proportional by uniqueness. Here the unknowns are in a module, as in Sharpe's paper, and the identity on which everything rests (Lemma 3) is proved for all coordinates at once by a contraction. The theorem is checked in Unit (`sag/depth2`).

1. Setting

Let $e_1, \ldots, e_m$ be the canonical basis of R^m. For a finite set $I = \{i_1 < \cdots < i_k\}$ write $e_I = e_{i_1} \wedge \cdots \wedge e_{i_k}$, and for x in the exterior algebra $\Lambda = \Lambda(R^m)$ write $[x]_I$ for its coordinate on e_I. Let

$$\omega = U_1 \wedge \cdots \wedge U_n, \qquad P_k(b) = U_1 \wedge \cdots \wedge U_{k-1} \wedge b \wedge U_{k+1} \wedge \cdots \wedge U_n \quad (b \in R^m).$$

The coordinates $\delta_I = [\omega]_I$ are the $n \times n$ minors of A, so $\Delta_n(A)$ is generated by the family $(\delta_I)_I$. For $F = R$ the coordinates of $\omega \wedge V$ are the $(n+1)$-minors of B; in general we put $\omega \wedge V = \sum_t (\omega \wedge e_t) \otimes V_t$, and its coordinates

$$\mu_I = \textstyle\sum_t [\omega \wedge e_t]_I\, V_t \in F$$

generate $\Delta_{n+1}(B)$. Cramer's numerators for the subsystem of the rows I are

$$\lambda_k(I) = \textstyle\sum_t [P_k(e_t)]_I\, V_t \in F,$$

which for $F = R$ is $[P_k(V)]_I$: in coordinates, $\delta_I e_I = U_1(I) \wedge \cdots \wedge U_n(I)$ and $\lambda_k(I) e_I = U_1(I) \wedge \cdots \wedge V(I) \wedge \cdots \wedge U_n(I)$, with $V(I)$ in place k and $X(I) = \sum_{i \in I} X_i e_i$ [2].

Grade is used in an explicit form [1, 3], equivalent for finitely generated ideals to the true grade of Hochster and Northcott [4]. For a finite family (d_I) of elements of R: $\mathrm{Gr}(d; F) \geq 1$ if $d_I x = 0$ for all I implies $x = 0$ ($x \in F$); and $\mathrm{Gr}(d; F) \geq 2$ if moreover every family (y_I) in F with $d_I y_J = d_J y_I$ for all I, J is of the form $y_I = d_I e$ with $e \in F$ (unique, by $\mathrm{Gr} \geq 1$).

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2. Two lemmas

Lemma 2 (Cramer)**.** *For $x \in F^n$ and every I and k, $\sum_t [P_k(e_t)]_I \, (\sum_j x_j U_j)_t = \delta_I \, x_k$. Hence, if $\mathrm{Gr}(\Delta_n(A); F) \geq 1$, the system (1) has at most one solution.*

Proof. For $F = R$ the left-hand side is $[P_k(\sum_j x_j U_j)]_I$, since P_k is linear, and $P_k(\sum_j x_j U_j) = \sum_j x_j P_k(U_j) = x_k \, \omega$, because $P_k(U_j)$ has the factor U_j twice for $j \neq k$. The general case is the same computation with coefficients in F. If x and y are solutions, then $\delta_I x_k = \delta_I y_k$ for all I, so $x = y$. □

Lemma 3 (key identity)**.** *Let φ be a linear form on R^m and ι_φ the contraction by φ on Λ. For every $b \in R^m$,*

$$\sum_j \varphi(U_j) \, P_j(b) = \varphi(b) \, \omega + (-1)^{n-1} \iota_\varphi(\omega \wedge b).$$

In particular, if $\omega \wedge b = 0$, then $\sum_j U_j^r \, [P_j(b)]_I = b_r \, \delta_I$ for every row r and every I.

Proof. ι_φ is a derivation of degree -1: $\iota_\varphi(w_1 \wedge \cdots \wedge w_k) = \sum_l (-1)^{l-1} \varphi(w_l) \, w_1 \wedge \ldots \widehat{w_l} \cdots \wedge w_k$. Applied to $U_1 \wedge \cdots \wedge U_n \wedge b$, the term $l = n+1$ is $(-1)^n \varphi(b) \, \omega$, and for $l = j \leq n$ moving b from the end to place j gives $U_1 \wedge \ldots \widehat{U_j} \cdots \wedge U_n \wedge b = (-1)^{n-j} P_j(b)$, so the term is $(-1)^{n-1} \varphi(U_j) P_j(b)$. For the second claim take $\varphi(w) = w_r$ and the coordinate on e_I. □

For $F = R$ the second claim is the identity proved in the appendix of [2] for $i \notin I$, by expanding $(V(i) + V(I)) \wedge (U_1(i) + U_1(I)) \wedge \cdots \wedge (U_n(i) + U_n(I)) = 0$; the contraction gives it for all I and r at once.

3. Proof of the theorem

Applying Lemma 3 to $b = e_t$, multiplying by V_t and summing over t gives, for every I and r,

$$\sum_j U_j^r \, \lambda_j(I) = \delta_I \, V_r + \sum_{I'} k_{I'} \, \mu_{I'}, \tag{2}$$

where $k_{I'} = (-1)^{n-1} [\iota_\varphi(e_{I'})]_I$, φ the r-th coordinate: the last term of Lemma 3 is linear in $\omega \wedge e_t$, hence a combination of its coordinates.

Existence. Assume $\Delta_{n+1}(B) = 0$, i.e. all $\mu_I = 0$. By (2), $\lambda(I) = (\lambda_1(I), \ldots, \lambda_n(I))$ satisfies

$$\sum_j \lambda_j(I) \, U_j = \delta_I \, V \qquad \text{for every } I.$$

For two subsets I, J the vectors $\delta_J \lambda(I)$ and $\delta_I \lambda(J)$ both solve the system with right-hand side $\delta_I \delta_J V$, so they are equal by Lemma 2: $\delta_J \lambda_k(I) = \delta_I \lambda_k(J)$ for every k. Since $\mathrm{Gr}(\Delta_n(A); F) \geq 2$, there is $x_k \in F$ with $\lambda_k(I) = \delta_I x_k$ for all I. Then

$$\delta_I \sum_j x_j U_j = \sum_j \lambda_j(I) \, U_j = \delta_I \, V$$

for all I, and $\sum_j x_j U_j = V$ since $\Delta_n(A)$ is regular on F. Uniqueness is Lemma 2.

Converse. If $\sum_j x_j U_j = V$, then $\mu_I = \sum_j x_j \, [\omega \wedge U_j]_I = 0$, since $\omega \wedge U_j$ has the factor U_j twice. □

4. The formalisation

The theorem is checked in Unit, in `sag/depth2`, with univalence and replacement as parameters as in the rest of `sag` (`./segc sag/depth2 all`: 170 modules, about 2.8 s, every file under 0.15 s). It uses the exterior algebra of `sag/exterior` (products of generators, coordinates, the contraction c_v by a vector and the grade involution α) and the convention of `sag/cffr` that minors are exterior coordinates. Grade is the explicit form above (`Depth`); for a family indexed by $\{1, \ldots, N\}$ it coincides with the Koszul grade of `sag/koszul` (`DepthKoszul`).

Lemma 3 is proved without signs $(-1)^j$, by induction on n in the form $c_\varphi(\omega) \wedge b = -\alpha\big(\sum_j \varphi(U_j)P_j(b)\big)$, using $c_\varphi(x \wedge w) = c_\varphi(x) \wedge w + \varphi(w)\,\alpha(x)$ and the anticommutation of generators; then $\sum_j \varphi(U_j)P_j(b) = \varphi(b)\,\omega - \alpha(c_\varphi(\omega \wedge b))$. The coefficients $k_{I'}$ in (2) come from expanding the linear form $y \mapsto [-\alpha(c_\varphi(y))]_I$ in the basis $(e_{I'})$.

`Depth`	$\mathrm{Gr} \geq 1$, $\mathrm{Gr} \geq 2$, uniqueness of the multiple
`WedgeContraction`	Lemma 3 (`derivation`, `General`, `coeff-identity`)
`SharpeExterior`	Lemma 2 (`pair`, `Inj`), (2) (`Key`), Theorem 1 (`System.theorem`, `sol-prop`, `necessary`); for $F = R$ with the hypothesis $\omega \wedge V = 0$ (`InRing.theorem`, and `lam-ring`: $\lambda_k(I) = [P_k(V)]_I$)

The same directory contains the other results of [1, B-7] in depth 2: the local-global principles for divisibility and strong gcds (B-7.1), for depth 2 itself (B-7.2 with $k = 2$), and the covering principle (B-7.4), for localisations at single elements.

References

[1] T. Coquand, H. Lombardi, *Résolutions libres finies. Méthodes constructives*, Calvage & Mounet, 2024.

[2] T. Coquand, *Grade and linear equations*, note, September 2010.

[3] T. Coquand, C. Quitté, *Constructive finite free resolutions*, manuscripta math. 137 (2012), 331–345.

[4] D. G. Northcott, *Finite Free Resolutions*, Cambridge Tracts in Mathematics 71, Cambridge University Press, 1976.

[5] D. W. Sharpe, *Grade and the theory of linear equations*, Linear Algebra Appl. 18 (1977), 25–32.